Maastricht University

Department of Advanced Computing Sciences

Queueing Theory

Barbara Franci

Queueing Processes

Call center:

- 30 calls per hour
- 3 agents reply and take 3 minutes per call (20 call per hour)
- If the agents are busy, costumers wait (up to 2)
- Any other call is lost, the costumers have to call another time

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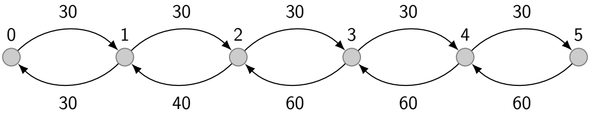

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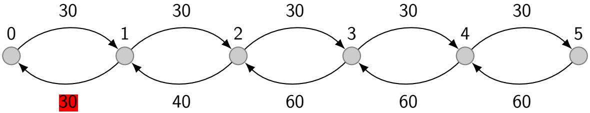

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This is a Markov Chain!

Queueing Processes

Interarrival time: time between calls
Service time: time to answer the call, i.e., how often a costumer leaves

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Goal:

- What fraction of time do the costumers spend in each state?
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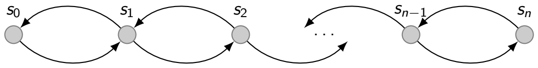
- How much time does it take to go from 0 to 3?

Relation to Markov Chains



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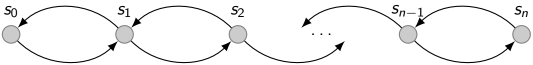
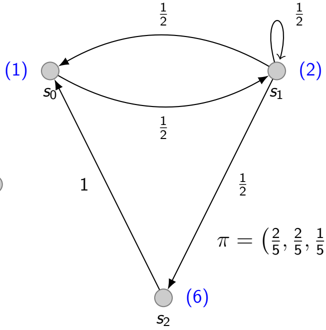


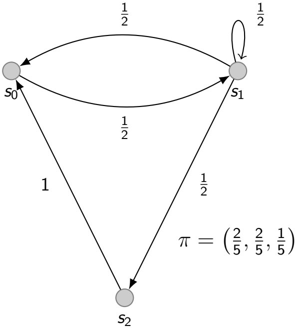


$\pi = \left(\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\right)$

Problems

- Queues involve waiting...

Parenthesis



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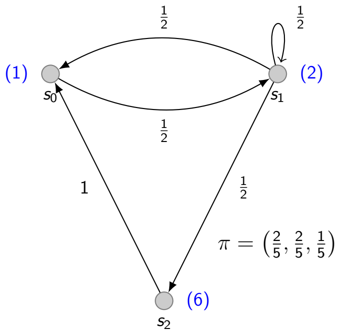

Let's consider the waiting time:

$$1 \cdot \frac{2}{5} + 2 \cdot \frac{2}{5} + 6 \cdot \frac{1}{5} = \frac{12}{5}$$

We have 12 time units to distribute among the states:

$\Rightarrow \pi = \left(\frac{2}{12}, \frac{4}{12}, \frac{6}{12}\right) = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$

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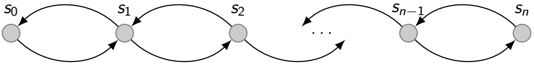

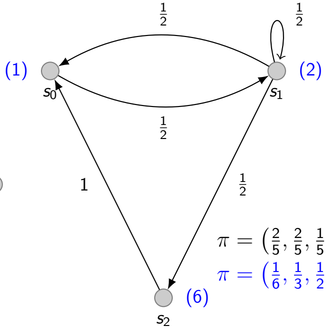

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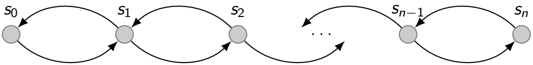

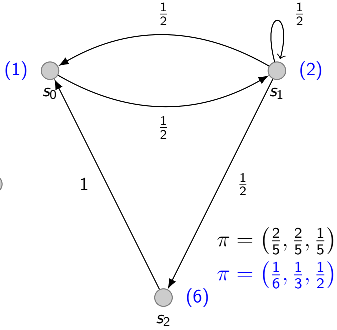

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- Costumers do not arrive uniformly
- Costumers are not processes uniformly

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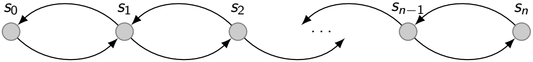

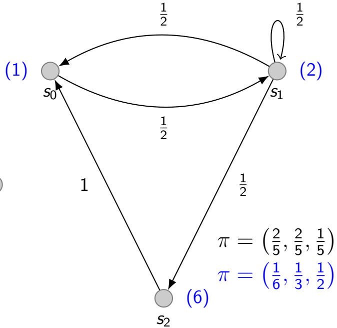

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- Costumers do not arrive uniformly
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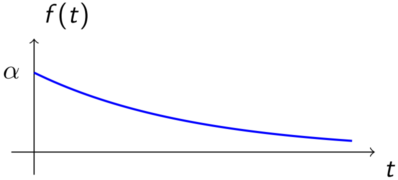
$\Rightarrow$ (something like) a Markov Chain with arrival/service times, according to a probability distribution
But Markov Chains need a kind of memory-less distribution...

Exponential Distribution

Let X be a random variable, e.g., for interarrival or service time. X has an exponential distribution with fixed parameter $\alpha > 0$ if the density function is

$$f(t) = \begin{cases} \alpha e^{-\alpha t} & t \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$$



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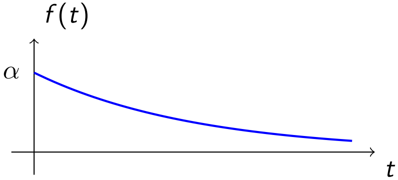

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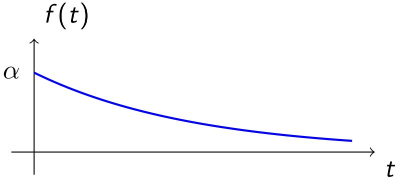

What is the expected time between arrivals?

$$\mathbb{E}[x] = \int_0^\infty t f(t) dt = \int_0^\infty t\alpha e^{-\alpha t} dt = -\ t e^{-\alpha t}\Big|_0^\infty - \int_0^\infty -e^{-\alpha t} dt$$
$$= -te^{-\alpha t} + \frac{1}{\alpha} e^{-\alpha t}\Big|_0^\infty = \frac{1}{\alpha}$$

If α costumers arrive in 1 hour, then they arrive one every $\frac{1}{\alpha}$ time

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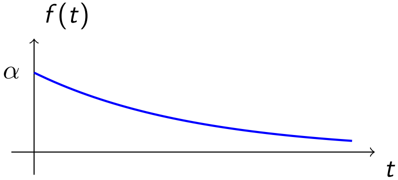

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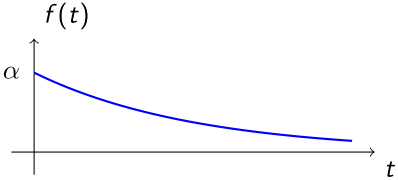

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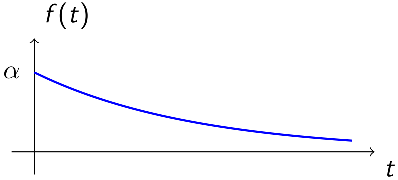

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$$\mathbb{P}[a < x < b] = \int_a^b \alpha e^{-\alpha t} dt = - \left. e^{-\alpha t} \right|_a^b = e^{-\alpha a} - e^{-\alpha b}$$

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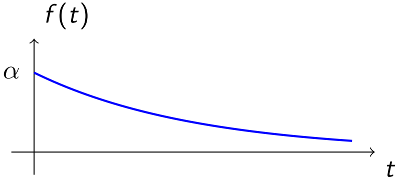

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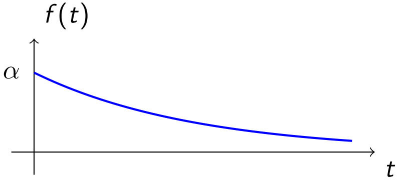

What is the probability of something happening given that something has already happened?

$$\begin{aligned} \mathbb{P}[x > a + c \mid x > c] &= \frac{\mathbb{P}[x > a + c \text{ and } x > c]}{\mathbb{P}[x > c]} = \frac{\mathbb{P}[x > a + c]}{\mathbb{P}[x > c]} \\ &= \frac{e^{-\alpha(a+c)}}{e^{-\alpha c}} = e^{-\alpha a} = \mathbb{P}[x > a] \end{aligned}$$

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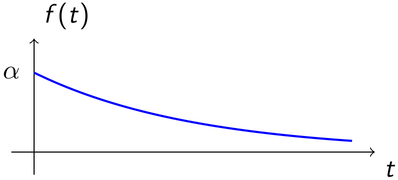

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Lack of memory: the probability of a new arrival is always the same independently of how many events there have been before

More exponential distribution

X, Y : independent random vars. with $X \sim \exp(\alpha)$, $Y \sim \exp(\beta)$.

- Let $Z = \min\{X, Y\}$, *i.e.*, Z is the first event among $\{X, Y\}$
  Observe that $Z \sim \exp(\alpha + \beta)$

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Poisson Distribution

Let X be a random variable for an arrival with $X \sim exp(\alpha)$.

- Prob(n arrivals in next x time units) $\sim$ Poisson distribution
- Prob(n arrivals in next x time units)=Prob(n arrivals in the first x) [memoryless]

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&= \int_0^x \alpha(x-t)e^{-\alpha(x-t)}\alpha e^{-\alpha t} dt \\
&= \int_0^x \alpha^2 (x-t) e^{-\alpha x} dt \\
&= \alpha^2 \left(xt - \frac{1}{2}t^2\right) e^{-\alpha x}\Big|_0^x = \frac{\alpha^2 x^2}{2} e^{-\alpha x}
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&= \int_0^x \alpha^2 \frac{(x-t)^2}{2} e^{-\alpha(x-t)} \alpha e^{-\alpha t} dt \\
&= \int_0^x \alpha^3 \frac{(x-t)^2}{2} e^{-\alpha x} dt \\
&= \frac{\alpha^3}{6}(x-t)^3 \cdot e^{-\alpha x} \Big|_0^x = \frac{\alpha^3 x^3}{6} e^{-\alpha x}
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This look very similar to the queueing process!

Recap

Goal:

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What we have is:

- Number of arrivals (via Poisson Distribution)
- Memory-Less property (via Exponential Distribution)
- Time before event happening (via $Z = \min\{X, Y\}$)
- Probability of event happening (via $\mathbb{P}(X < Y)$)

Is this enough?

An example

Medical Clinic with several types of patients, e.g., patients with flu symptoms arrive 10 /hour, and phyical injury patients arrive 5/hour. [both exp distributed]

- What are α, β, where flu $\sim$ exp(α) and injury $\sim$ exp(β) ?
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- How long until the first (i) flu; (ii) flu or injury patient arrives?
  (i) flu $\sim$ exp(10), thus $\mathbb{E}$ (first flu patient arrival) $= \frac{1}{10}$.
  (ii) arrival of flu OR injury $\sim$ min (flu, injury) $\sim$ exp($10 + 5$), thus

$$\mathbb{E}(\text{first patient arrival}) = \frac{1}{15} \quad \text{[inuitively, 15 expected events]}$$

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An example

Medical Clinic with several types of patients, e.g., patients with flu symptoms arrive 10 /hour, and phyical injury patients arrive 5/hour. [both exp distributed]

- What are α, β, where flu $\sim$ exp(α) and injury $\sim$ exp(β) ?

$$\alpha = 10, \beta = 5$$

- How long until the first (i) flu; (ii) flu or injury patient arrives?
  (i) flu $\sim$ exp(10), thus $\mathbb{E}$ (first flu patient arrival) $= \frac{1}{10}$.
  (ii) arrival of flu OR injury $\sim$ min (flu, injury) $\sim$ exp($10 + 5$), thus

$$\mathbb{E}(\text{first patient arrival}) = \frac{1}{15} \quad \text{[inuitively, 15 expected events]}$$

- What is $\mathbb{P}$(first patient has flu symptoms)?

$$\mathbb{P}(\text{first patient has flu symptoms}) = \frac{\alpha}{\alpha + \beta} = \frac{10}{15}$$

Example

Call Center

- 30 calls per hour exponentially distributed, i.e., $exp(30)$
- 2 agents reply and can serve 20 costumers per hour, i.e., $exp(20)$
- If the agents are busy, costumers wait (up to 2)
- Any other call is lost, the costumers have to call another time

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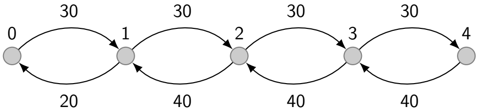

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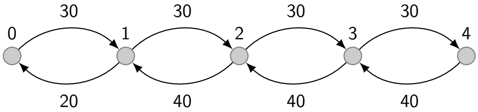

- What is the probability of leaving state s_3 for s_4?
- How much time do we spend in state s_2?

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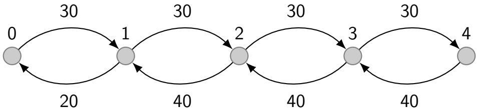

- What is the probability of leaving state s_3 for s_4? $\frac{30}{30+40}$
- How much time do we spend in state s_2? $\frac{1}{30+40}$

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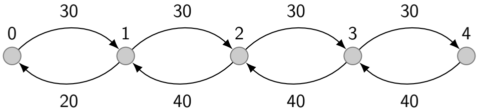


How many events/per hour cause us to move from state s_i to state s_j ?

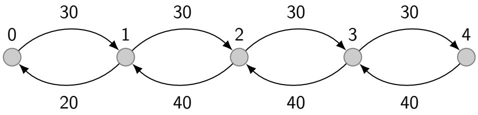
$s_i \rightarrow s_{i+1}$: a customer arrived.

- 30 customers/hour for all i

$s_{i+1} \rightarrow s_i$: a customer departed.

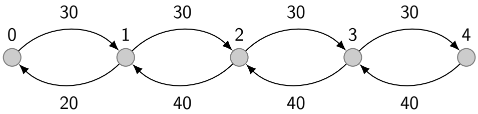
- 20 customers/hour ($i = 1$)
- 40 customers/hour ($i > 1$)

Example



What fraction of time do the costumers spend in each state?

Example

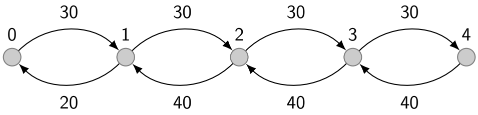


What fraction of time do the costumers spend in each state?

$$
\begin{aligned}
\text{In} &= \text{Out} \\
20\pi_1 &= 30\pi_0 \\
30\pi_0 + 40\pi_2 &= 20\pi_1 + 30\pi_1 \\
30\pi_1 + 40\pi_3 &= 40\pi_2 + 30\pi_2 \\
30\pi_2 + 40\pi_4 &= 40\pi_3 + 30\pi_3 \\
30\pi_3 &= 40\pi_4
\end{aligned}
$$

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

Example



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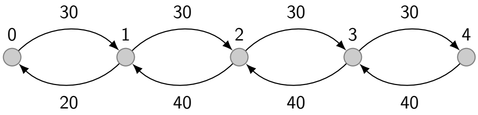
$$\begin{aligned} \text{In} &= \text{Out} \\ 20\pi_1 &= 30\pi_0 \\ 40\pi_2 &= 30\pi_1 \\ 40\pi_3 &= 30\pi_2 \\ 40\pi_4 &= 30\pi_3 \\ 30\pi_3 &= 40\pi_4 \end{aligned}$$

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

$$\pi = \frac{1}{653}(128, 192, 144, 108, 81)$$

Example



What is the average number of costumers in the system?

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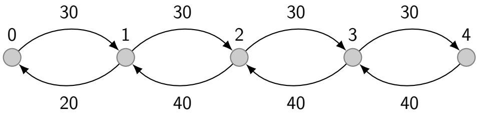

What is the average number of costumers in the system?

$$\pi = \frac{1}{653}(128, 192, 144, 108, 81)$$

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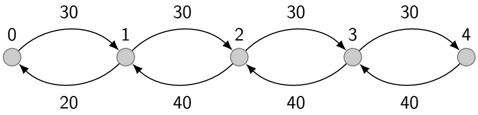
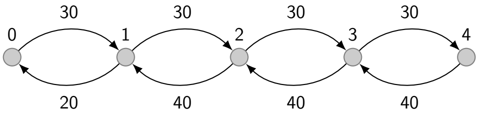


What is the average number of costumers in the system?

$$\pi = \frac{1}{653}(128, 192, 144, 108, 81)$$

$$\text{Avg. Num. Costumers} = 0 \cdot \pi_0 + 1 \cdot \pi_1 + 2 \cdot \pi_2 + 3 \cdot \pi_3 + 4 \cdot \pi_4$$

Example

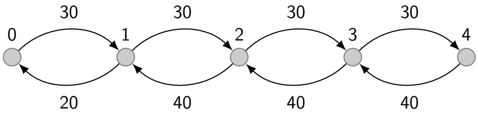


What is the average number of costumers in the system?

$$\pi = \frac{1}{653}(128, 192, 144, 108, 81)$$

$$\begin{aligned}
\text{Avg. Num. Costumers} &= 0 \cdot \pi_0 + 1 \cdot \pi_1 + 2 \cdot \pi_2 + 3 \cdot \pi_3 + 4 \cdot \pi_4 \\
&= \left(0 \cdot \frac{128}{653} + 1 \cdot \frac{192}{653} + 2 \cdot \frac{144}{653} + 3 \cdot \frac{108}{653} + 4 \cdot \frac{81}{653} \right) \\
&= \frac{1128}{653} \sim 1.7
\end{aligned}$$

Example



How much time does it take to go from 0 to 3?

$$
\begin{aligned}
E_x &= \text{ expected time until there are 3 customers when starting in } x \\
&= \text{ expected time to go from } s_x \text{ to } s_3 \\
&= \text{ Time in } x + \sum P_{xi} E_i
\end{aligned}
$$

- How much time in x?
- What are the transition probabilities P_{xi}?

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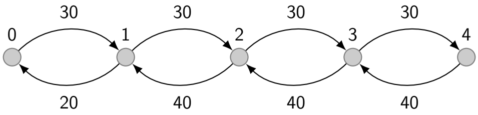

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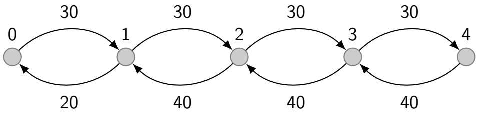

- How much time in x?
  - 30 events/hour leave $s_0 \Rightarrow \frac{1}{30}$ hours/visit to s_0
  - $20 + 30$ events/hour leave $s_1 \Rightarrow \frac{1}{50}$ hours/visit to s_1
  - 70 events/hour leave $s_i, i \in \{2, \ldots, w+1\} \quad \Rightarrow \frac{1}{70}$ hours/visit to s_i
  - 40 events/hour leave $s_{w+2} \Rightarrow \frac{1}{40}$ hours/visit to s_{w+2}
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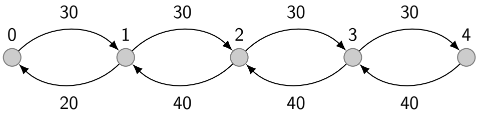

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  - in s2, leaving to s1 occurs with probability $\frac{40}{40+30}$
  - in s2, leaving to s3 occurs with probability $\frac{30}{40+30}$

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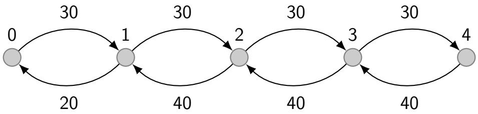

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$\Rightarrow$ Set up the system and solve:

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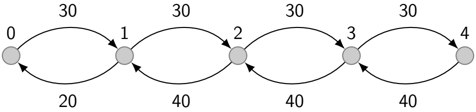

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$$
\begin{aligned}
E_0 &= \frac{1}{30} + E_1 \\
E_1 &= \frac{1}{50} + \frac{2}{5}E_0 + \frac{3}{5}E_2 \\
E_2 &= \frac{1}{70} + \frac{4}{7}E_1 + \frac{3}{7}E_3 \\
E_3 &= 0
\end{aligned}
$$

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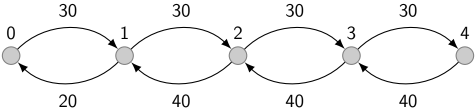

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$$E_0 = \frac{53}{270} \text{ hours}$$

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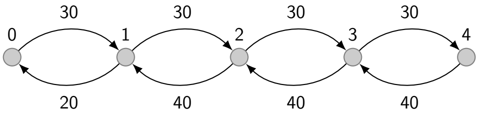

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$$E_0 = 2 + E_1$$
$$E_1 = \frac{6}{5} + \frac{2}{5}E_0 + \frac{3}{5}E_2$$
$$E_2 = \frac{6}{7} + \frac{4}{7}E_1 + \frac{3}{7}E_3$$
$$E_3 = 0$$

$$E_0 = \frac{53}{270} \text{ hours}$$

$$E_0 = \frac{106}{9} \sim 11,8 \text{ minutes}$$